

B. Tech Degree V Semester Examination, November 2008

ME 505 HEAT AND MASS TRANSFER

(2002 Scheme)

Time : 3 Hours

Maximum Marks : 100

(Data book may be permitted)

- I. (a) Show that the temperature distribution in a sphere of radius R , having a surface temperature of T_w in which heat is generated at a uniform rate of q_g is given by
- $$T_r - T_w = \frac{q_g R^2}{6K} \left(1 - \frac{r^2}{R^2} \right) \text{ where } K \text{ is the thermal conductivity of the sphere material.} \quad (10)$$
- (b) A layer of 5 cm thick insulating brick having conductivity of 1.5 W/mk is placed between two 0.5 cm thick steel plates. The conductivity of mild steel is 50 W/mk. The faces of brick adjacent to the plates are rough having solid to solid contact of 30% of the total area. The average height of the asperities is 0.1 cm. If the outer plate surface temperatures are 100°C and 500°C respectively, calculate the rate of heat transfer per unit area. The conductivity of air is 0.02 W/mk. (10)
- OR**
- II. (a) Derive an equation for temperature distribution in a fin with an insulated tip. Then obtain an expression for heat dissipation by integrating the convective losses along it's surface. (10)
- (b) A 12 mm diameter mild steel sphere ($K = 42.5$ W/mk) is exposed to cooling airflow at 27°C resulting in the convective co-efficient $h = 114$ W/m²K. Determine
- (i) time required to cool the sphere from 540°C to 95°C
 - (ii) instantaneous heat transfer rate 2 minutes after the start of cooling
 - (iii) total energy transferred from the sphere during the first 2 minutes. (10)
- III. (a) Compare the variations of velocity, temperature and local heat transfer co-efficient along a vertical plate for the plate under natural convection and forced convection. (10)
- (b) Air at 25°C flows over a thin plate with a velocity of 2.5 m/s. The plate is 2 m long and 1 m wide. Estimate the thermal boundary layer thickness at the trailing edge of the plate and total drag force experienced by the plate. At 25°C, the density of air is 1.2 Kg/m³ and kinematic viscosity is 15×10^{-6} m²/s. Prandtl number for air is 0.69. (10)
- OR**
- IV. (a) What is Reynolds analogy? Describe the relation between fluid friction and heat transfer. (8)
- (b) Water is boiled at a rate of 30 Kg/h in a copper pan, 30 cm in diameter, at atmospheric pressure. Estimate the temperature of the bottom surface of the pan assuming nucleate boiling conditions. (12)
- V. (a) Explain electrical network analogy for thermal radiation systems. (10)
- (b) The net radiation from the surface of two parallel plates maintained at temperatures, T_1 and T_2 is to be reduced by 79 times. Calculate the number of screens to be placed between the two surfaces to achieve this reduction in heat exchange, assuming the emissivity of the screens are 0.05 and that of the surfaces as 0.8. (10)

OR

(Turn Over)

- VI. (a) Explain the meaning of the term geometric shape factor in relation to heat exchange by radiation. Derive an expression for geometric factor F_{11} for the inside surface of a black hemispherical cavity of radius R with respect to itself. (6)
- (b) Two parallel plates of size 1 m by 1 m spaced 0.5 m apart are located in a very large room, the walls of which are maintained at a temperature of 27°C. One plate is maintained at a temperature of 900°C and the other at 400°C. Their emissivities are 0.2 and 0.5 respectively. If the plates exchange heat between themselves and surroundings, find the net heat transfer to each plate and to the room. Consider only the plate surfaces facing each other. (14)

- VII. (a) For analysis of heat exchanger problems, in which cases NTU method is applied. (5)
- (b) In a double pipe heat exchanger $M_h C_h = 0.5 M_c C_c$, M denotes mass flow rate (Kg/s), C denotes specific heat (J/KgK) and suffix h and c denotes hot and cold fluid respectively. The hot and cold fluids enter the heat exchanger at temperatures T_{h1} and T_{c1} respectively. Deduce an expression, in terms of T_{h1} , T_{c1} and T_{h2} , for the ratio of the area of a counter flow heat exchanger to that of a parallel flow heat exchanger which will give the same hot fluid outlet temperature T_{h2} . What is the ratio if $T_{h1} = 150^\circ\text{C}$, $T_{c1} = 30^\circ\text{C}$ and $T_{h2} = 90^\circ\text{C}$? (15)

OR

- VIII. In a counter flow double pipe heat exchanger water flows through a copper tube (19 mm OD and 16 mm ID), at a flow rate of 1.48 m/s. The oil flows through the annulus formed by inner copper tube and outer steel tube (30 mm OD and 26 mm ID). The steel tube is insulated from outside. The oil enters at 0.4 Kg/s and is cooled from 65°C to 50°C where as water enters at 32°C. Neglecting the copper tube, calculate the length of tube required. Data given :
- $N_u = 0.023 (Re)^{0.8} (Pr)^{0.3}$
- | | |
|----------------------------|---------------------------|
| Fouling factor, water side | 0.0005 m ² K/W |
| Fouling factor, oil side | 0.0008 m ² K/W |
- Water and oil properties :
- | | | |
|-----------------------------|-----------------------|-----------------------|
| Property | Oil | Water |
| ρ (Kg/m ³) | 850 | 995 |
| C_p (KJ/Kg K) | 1.89 | 4.187 |
| K (W/mK) | 0.138 | 0.615 |
| ν (m ² /s) | 7.44×10^{-6} | 4.18×10^{-7} |
- (20)

- IX. (a) Discuss the analogy between heat and mass transfer. (8)
- (b) CO₂ and air experience equimolar counter diffusion in a circular tube whose length and diameter are 1 m and 50 mm respectively. The system is at a total pressure of 1 atm and a temperature of 25°C. The ends of the tube are connected to large chambers in which the species concentrations are maintained at fixed values. The partial pressure of CO₂ at one end is 190 mm of Hg while at the other end is 95 mm of Hg. Estimate the mass transfer rate of CO₂ and air through the tube. (12)

OR

- X. (a) Derive an expression for the isothermal evaporation of water from a surface and it's subsequent diffusion through a stagnant air layer above. (10)
- (b) Estimate the diffusion rate of water at 27°C from the bottom of a test tube 20 mm is diameter and 4 cm long into dry air at 27°C. Take the diffusion co-efficient of water in air as $0.26 \times 10^{-4} \text{ m}^2/\text{s}$. (10)

